1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 synthesis highlights that tumors with visible TLS-like lymphoid aggregates do not always show improved outcomes — why do some TLS-rich tumors fail to deliver a survival benefit? The Ontologist's core paths tie a membrane transporter (SLC6A4) into a miRNA-regulated Wnt/β■catenin axis and show concurrent metabolic (glycolysis/HIF1A) and developmental (Hedgehog) programs that can reshape immune organization?

Working Hypothesis

Do TLS-like lymphoid aggregates fail to confer a survival advantage in tumors with high tumor-intrinsic SLC6A4 activity driving Wnt/β-catenin (CTNNB1) signaling, thereby altering TLS composition and function and suppressing anti-tumor immunity? If this mechanism operates, TLS-rich tumors with high SLC6A4/CTNNB1 activity will have worse overall survival than TLS-rich tumors with low activity, indicating a context-dependent decoupling of TLS presence from clinical benefit.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	SLC6A4, CTNNB1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

SLC6A4: SLC6A4 → miR-139-5p → Wnt/β■catenin) because it directly links a tumor-intrinsic signaling axis to β■catenin activity known to modulate immune architecture, and CORE PATH 2 (glycolysis) / CORE PATH 3 (Hedgehog) informed the broader context but were not included as core anchors to keep the hypothesis focused.

CTNNB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	miR-139-5p	Supporting modifier from knowledge graph
Secondary	CXCL12	Supporting modifier from knowledge graph

Secondary	CXCR4	Supporting modifier from knowledge graph
Secondary	HIF1A	Supporting modifier from knowledge graph
Secondary	glycolytic_genes	Supporting modifier from knowledge graph
Secondary	MYC	Supporting modifier from knowledge graph
Secondary	ECM_components	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: observational analyses in TCGA/multi-omics cohorts to correlate TLS presence/quality with SLC6A4/CTNNB1 activity signatures and OS, controlling for confounders;

Prediction 2: spatial transcriptomics and multiplex imaging to compare TLS microenvironments in TLS-rich tumors stratified by pathway activity;

Prediction 3: in vitro studies manipulating SLC6A4 and CTNNB1 in tumor cells to assess β-catenin activity, chemokine/antigen-presentation gene expression, and lymphocyte recruitment in co-culture or organoid systems;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- miR-139-5p: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CXCL12: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.